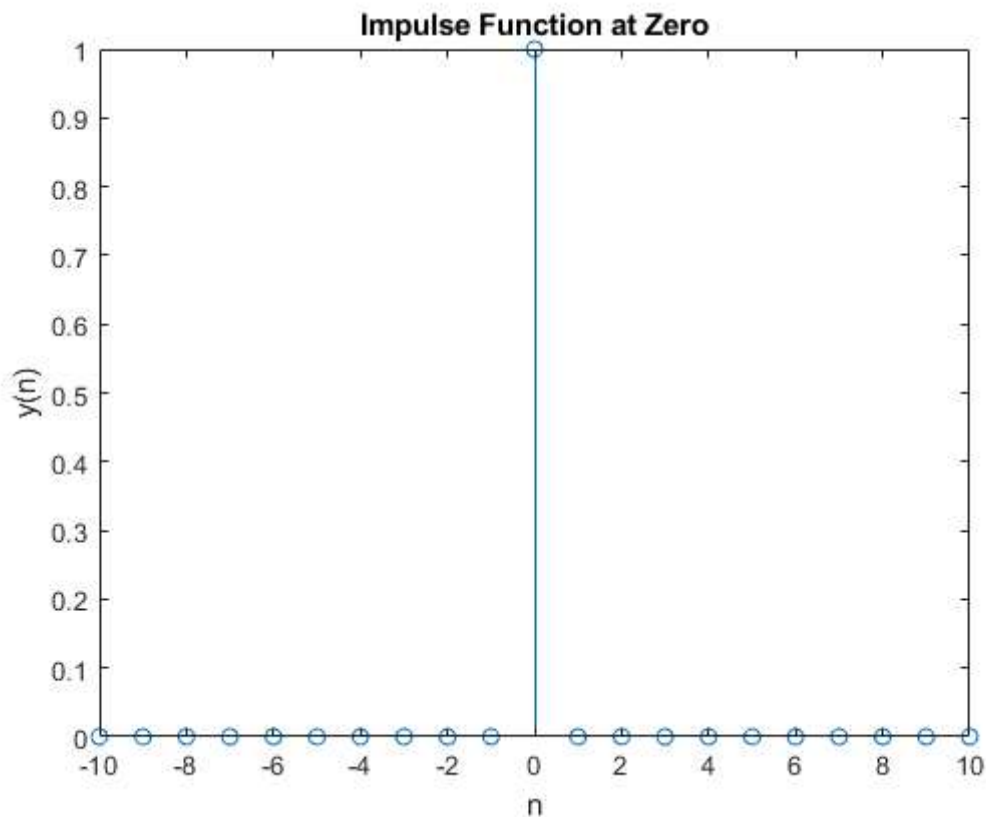


Problem 1 Plots

```
clear;  
[y, n] = unit_impulse(0, -10, 10)
```

```
y = 1×21  
    0    0    0    0    0    0    0    0    0    0    1    0    0    0    0    0  
n = 1×21  
   -10   -9   -8   -7   -6   -5   -4   -3   -2   -1    0    1    2    3    4    5
```

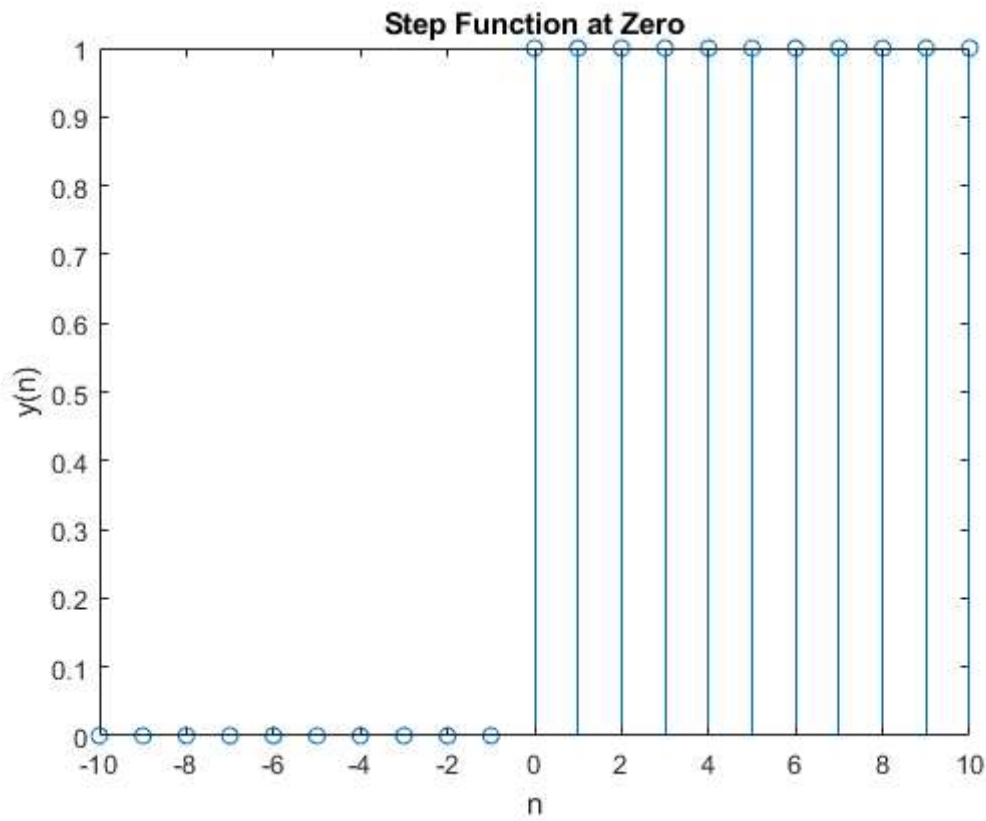
```
stem(n,y)  
title('Impulse Function at Zero');  
xlabel('n');  
ylabel('y(n)');
```



```
[y, n] = unit_step(0, -10, 10)
```

```
y = 1×21  
    0    0    0    0    0    0    0    0    0    0    1    1    1    1    1    1  
n = 1×21  
   -10   -9   -8   -7   -6   -5   -4   -3   -2   -1    0    1    2    3    4    5
```

```
stem(n, y)  
title('Step Function at Zero');  
xlabel('n');  
ylabel('y(n)');
```



Problem 2 Time Shift Plot

```
[y, n] = time_shift(y, n, 3)
```

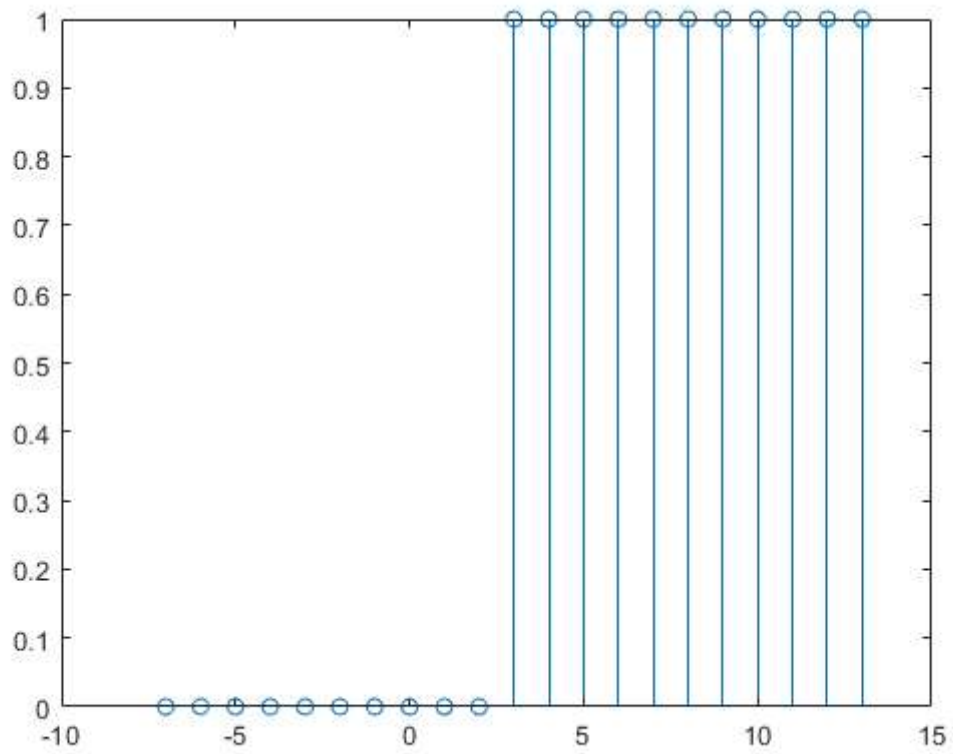
```
y = 1×21
```

```
0    0    0    0    0    0    0    0    0    0    1    1    1    1    1    1    1
```

```
n = 1×21
```

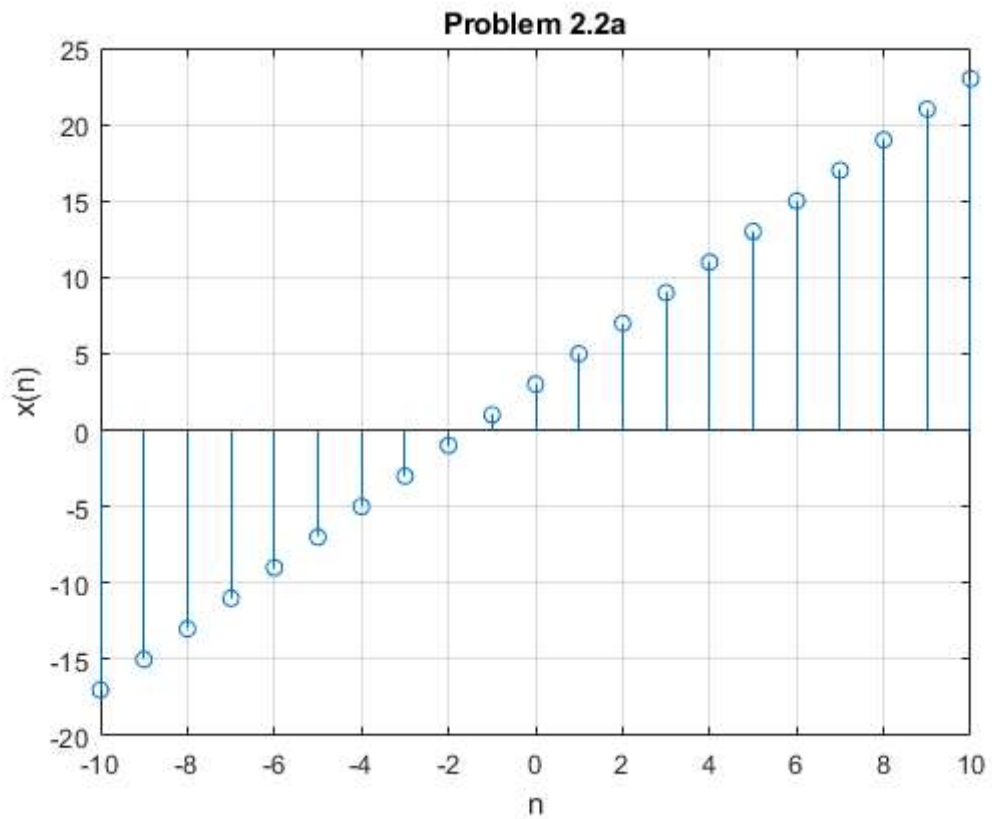
```
-7   -6   -5   -4   -3   -2   -1    0    1    2    3    4    5    6    7    8
```

```
stem(n, y)
```



Problem 2.2 Plots

```
clear;  
n = -10:10;  
x = (2*n)+3;  
stem(n, x)  
grid on;  
title('Problem 2.2a');  
xlabel('n');  
ylabel('x(n)');
```



```
[x_delay, n] = time_shift(x, n, 3)
```

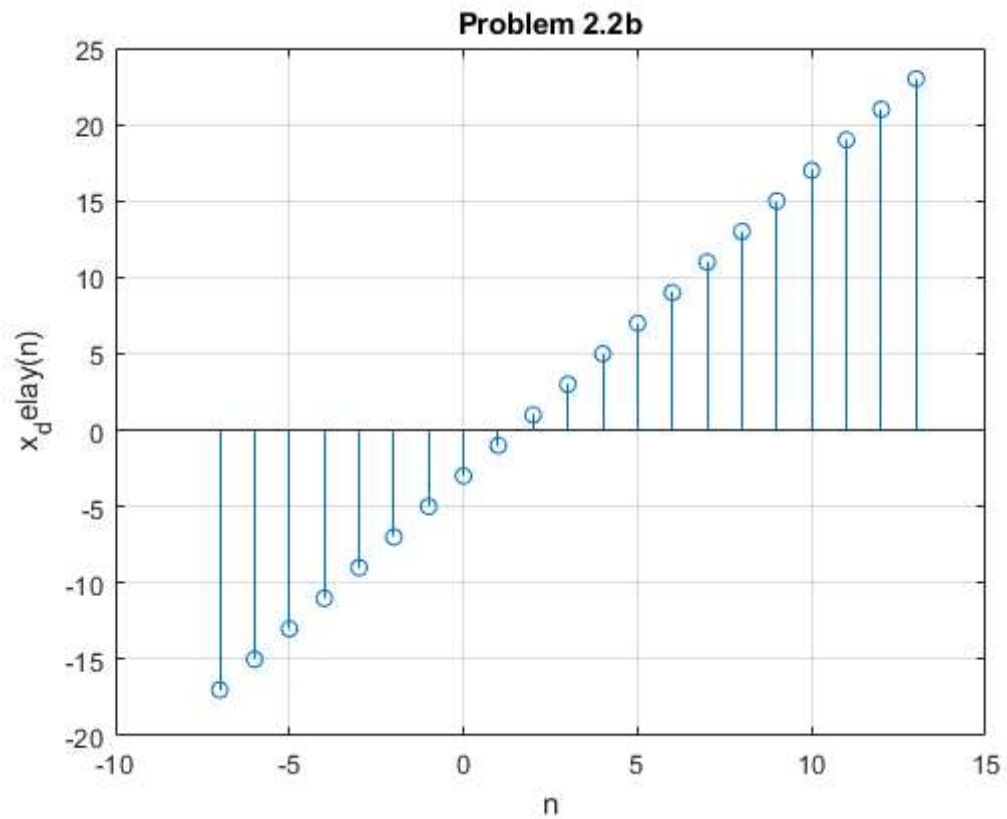
```
x_delay = 1×21
```

```
    -17    -15    -13    -11     -9     -7     -5     -3     -1     1     3     5     7     9    11    13
```

```
n = 1×21
```

```
     -7     -6     -5     -4     -3     -2     -1      0      1      2      3      4      5      6      7      8
```

```
stem(n, x_delay)
grid on;
title('Problem 2.2b');
xlabel('n');
ylabel('x_delay(n)');
```



```
n = -10:10
```

```
n = 1×21
```

```
-10    -9    -8    -7    -6    -5    -4    -3    -2    -1     0     1     2     3     4     5
```

```
[x_reverse, n] = time_reversal(x, n)
```

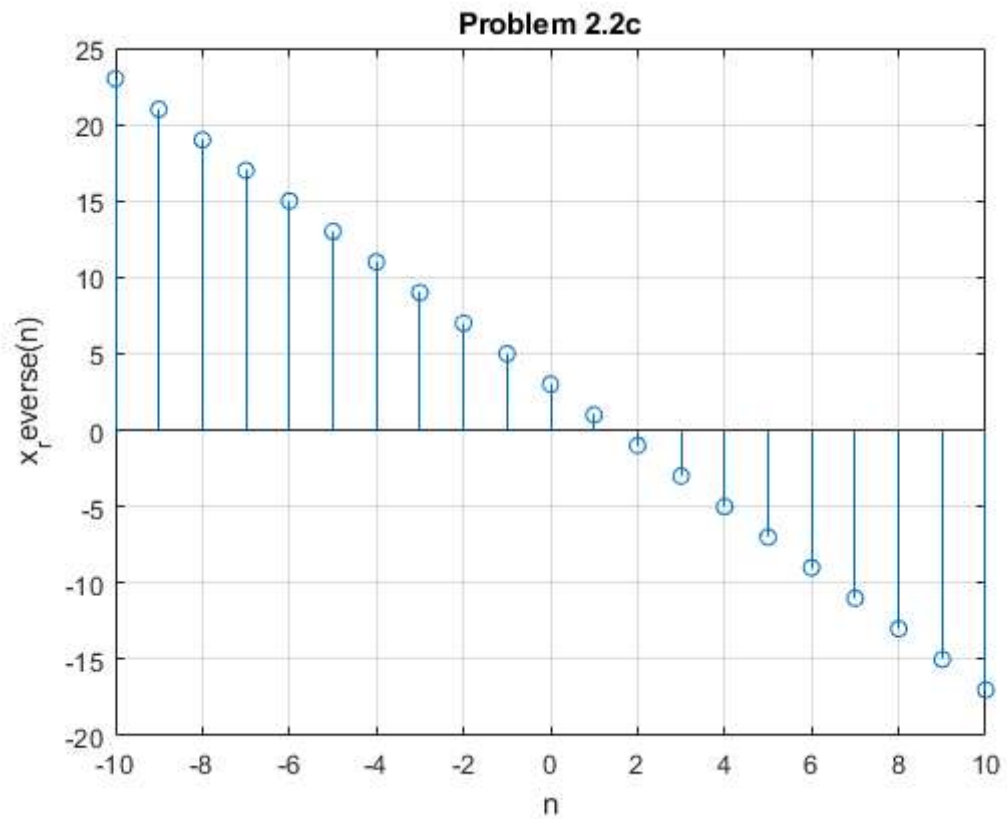
```
x_reverse = 1×21
```

```
23    21    19    17    15    13    11     9     7     5     3     1    -1    -3    -5    -7
```

```
n = 1×21
```

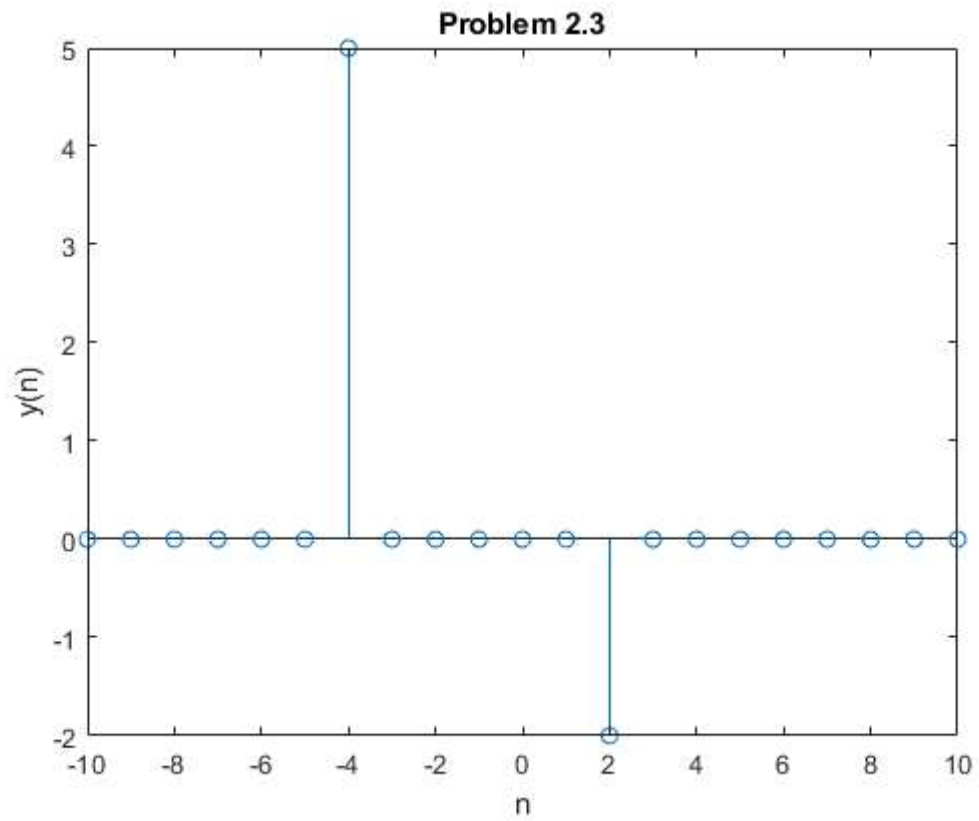
```
-10    -9    -8    -7    -6    -5    -4    -3    -2    -1     0     1     2     3     4     5
```

```
stem(n, x_reverse)
grid on;
title('Problem 2.2c');
xlabel('n');
ylabel('x_reverse(n)');
```



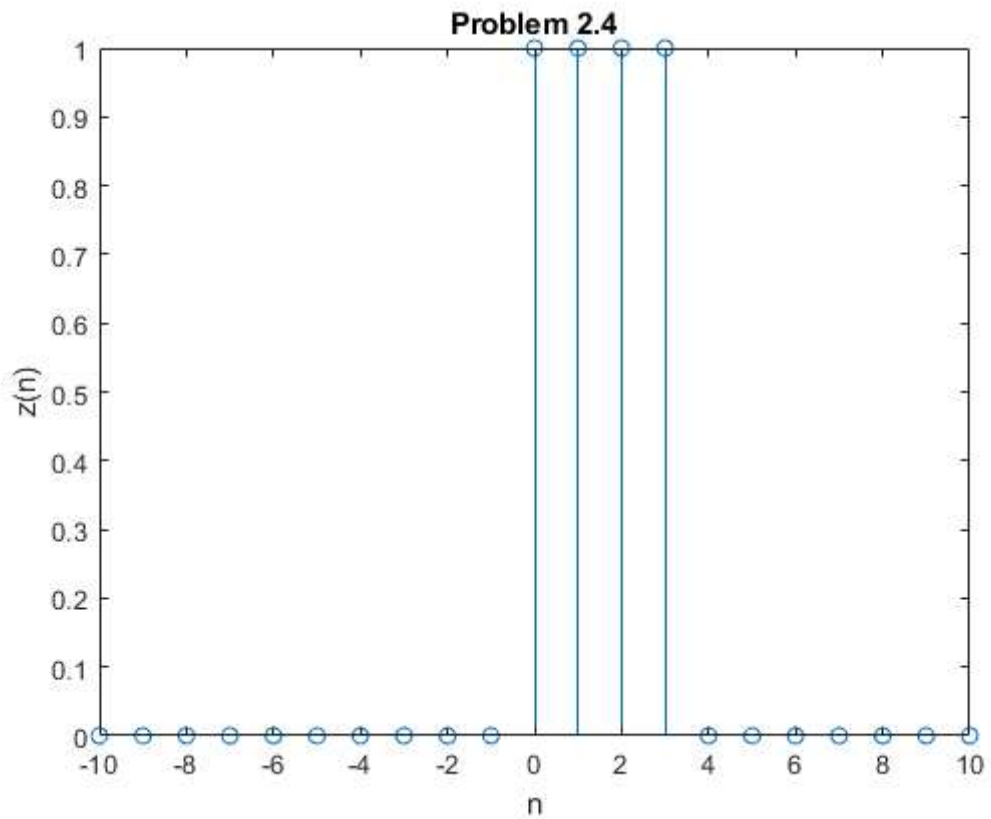
Problem 2.3

```
clear;
[y1, n] = unit_impulse(-4, -10, 10);
y1 = 5 .* y1;
[y2, n] = unit_impulse(2, -10, 10);
y2 = 2 .* y2;
y = y1 - y2;
stem(n, y)
title('Problem 2.3');
xlabel('n');
ylabel('y(n)');
```



Problem 2.4

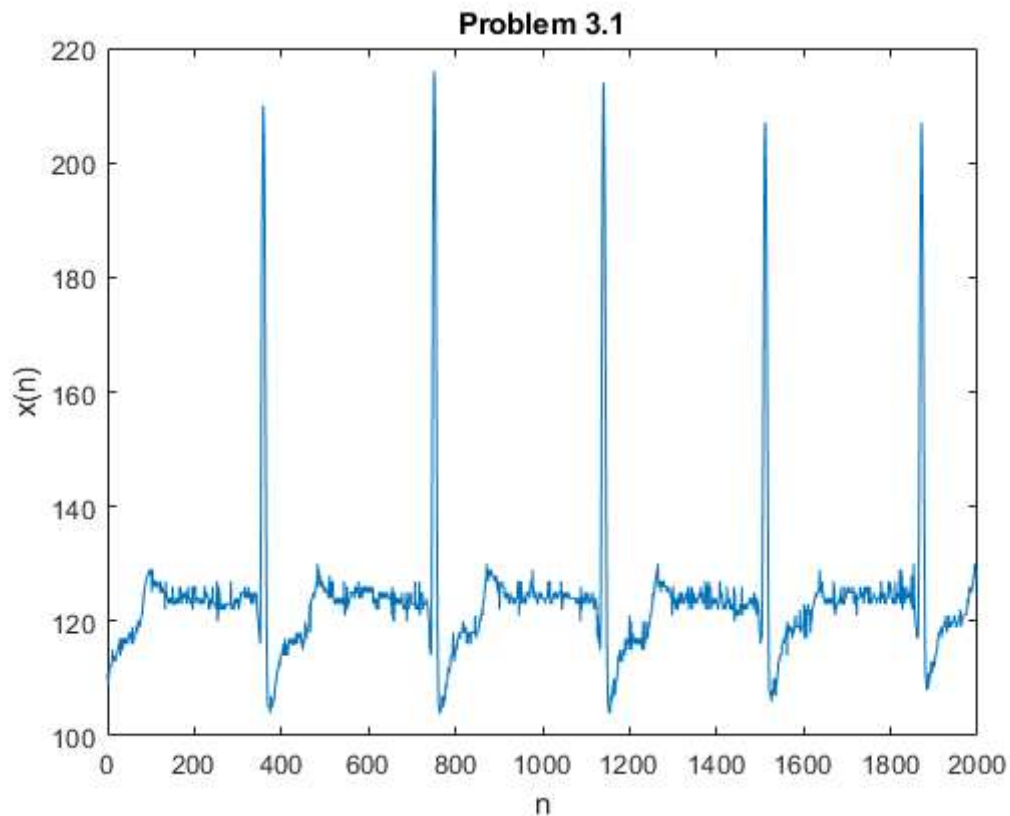
```
clear;
[y1, n] = unit_step(0, -10, 10);
[y2, n] = unit_step(4, -10, 10);
y = y1 - y2;
stem(n, y)
title('Problem 2.4');
xlabel('n');
ylabel('z(n)');
```



Problem 3

```
load('SAMPLE_ECG.mat')
x = ECG_Data;
n = 0:length(x)-1;

plot(n, x)
axis([0 2000 100 220]);
title('Problem 3.1');
xlabel('n');
ylabel('x(n)');
```



```

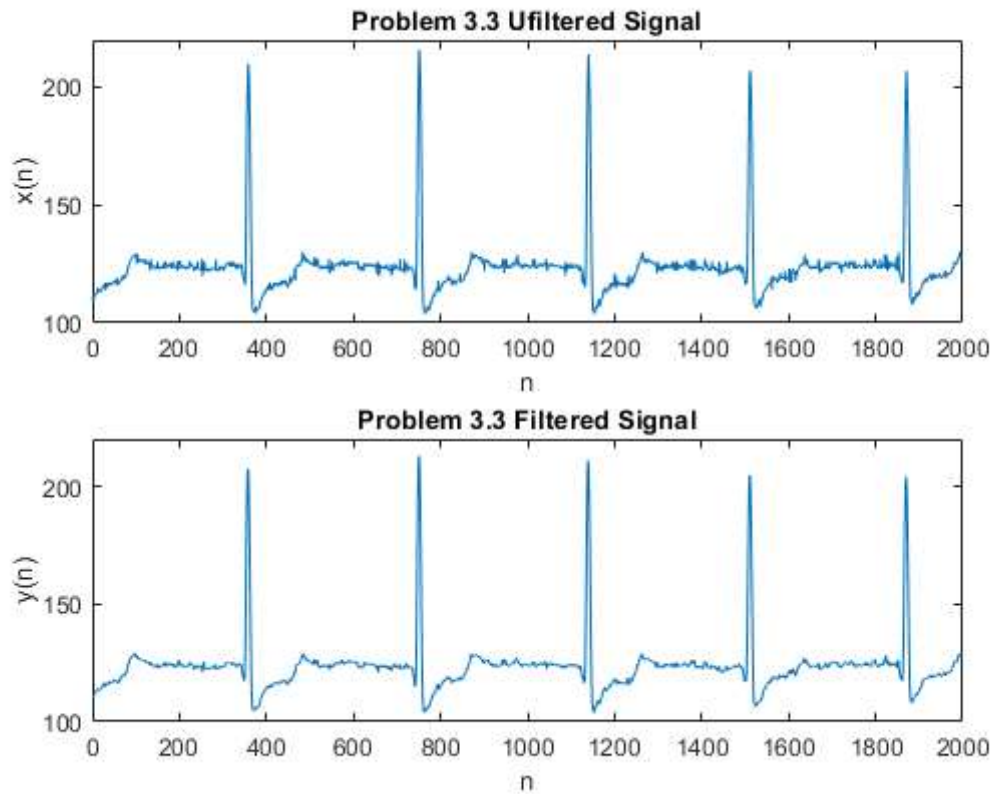
subplot(2, 1, 1)
plot(n, x)
axis([0 2000 100 220]);
title('Problem 3.3 Unfiltered Signal');
xlabel('n');
ylabel('x(n)');

for i = 1:length(x) - 2
    y(i) = (x(i) + x(i+1) + x(i+2)) / 3;
end

n = 0:length(x)-3;

subplot(2, 1, 2)
plot(n, y)
axis([0 2000 100 220]);
title('Problem 3.3 Filtered Signal');
xlabel('n');
ylabel('y(n)');

```



Problem 3.4

The signal $y(n)$ is much smoother than $x(n)$ and has much fewer visible noise and jumps. The filter effectively removed the high frequency noise.

Problem 1: Unit Impulse and Unit Step Functions

```
function [y,n] = unit_impulse(n_p, n1, n2)
    n = n1:n2;
    y = zeros(size(n));
    y(n == n_p) = 1;
end

function [y,n] = unit_step(n_p, n1, n2)
    n = n1:n2;
    y = zeros(size(n));
    y(n >= n_p) = 1;
end

function [y, n] = time_shift(x, n, n_d)
    n = n + n_d;
    y = x;
end

function [y, n] = time_reversal(x, n)
    y = fliplr(x);
    n = -fliplr(n);
end
```

